

Infinite dimensional Hilbert spaces

1. Discrete basis to Continuous basis
2. Coordinate basis representation
3. Dirac delta function

$$\textcircled{1} \quad \tilde{Q} = \tilde{H} = \begin{pmatrix} h & g \\ g & h \end{pmatrix} : h, g \in \mathbb{R}$$

$$\tilde{H}|\lambda\rangle = E|\lambda\rangle \quad E_0 = \sqrt{2}$$

$$= \frac{\beta}{\sqrt{2}-1} \frac{\sqrt{2}}{\sqrt{2}} \begin{pmatrix} 1+\sqrt{2} & 1 \\ 1 & -(1-\sqrt{2}) \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = 0 \sim$$

$$\tilde{A} \vec{x} = \lambda \vec{x}$$

$$\alpha = \frac{-\beta}{1-\sqrt{2}} \Leftrightarrow (1+\sqrt{2})\alpha + \beta = 0$$

$$\alpha - \beta(1-\sqrt{2}) = 0$$

$$E_0 = h+g$$

$$|0_H\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\alpha = \beta(1-\sqrt{2}) \quad E_0 = \sqrt{2}$$

$$|0_H\rangle = \frac{1}{\sqrt{(1+\sqrt{2})^2 + 1}} \begin{pmatrix} 1+\sqrt{2} \\ 1 \end{pmatrix}$$

$$E_1 = h-g$$

$$|1_H\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$|1_H\rangle = \frac{(1-\sqrt{2})|0\rangle + |1\rangle}{\sqrt{(1-\sqrt{2})^2 + 1}}$$

$$\textcircled{2} \quad \tilde{H} = E \begin{pmatrix} |0\rangle\langle 0| & -|1\rangle\langle 1| \\ +|0\rangle\langle 1| & +|1\rangle\langle 0| \end{pmatrix}$$

Reality check

$$\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} \frac{1+\sqrt{2}}{\sqrt{(1+\sqrt{2})^2 + 1}} \\ \frac{1+\sqrt{2}}{\sqrt{(1+\sqrt{2})^2 + 1}} \end{pmatrix}$$

$$\begin{pmatrix} \frac{1+\sqrt{2}}{\sqrt{(1+\sqrt{2})^2 + 1}} + 1 \\ \frac{1+\sqrt{2}}{\sqrt{(1+\sqrt{2})^2 + 1}} - 1 \end{pmatrix}$$

Stationary states

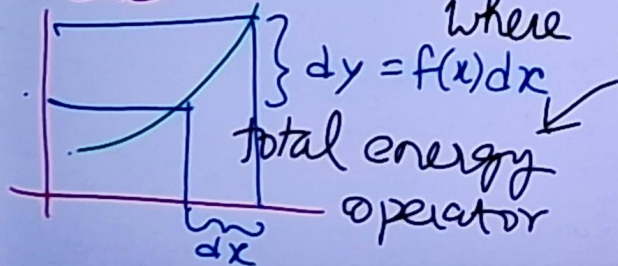
Deterministic states of Hamiltonian operator are stationary states.

Recall Postulate 3 of QM.

◉ (i) $|\psi\rangle$ evolves according to Schrödinger equation

Math

$$\frac{dy}{dx} = f(x)$$

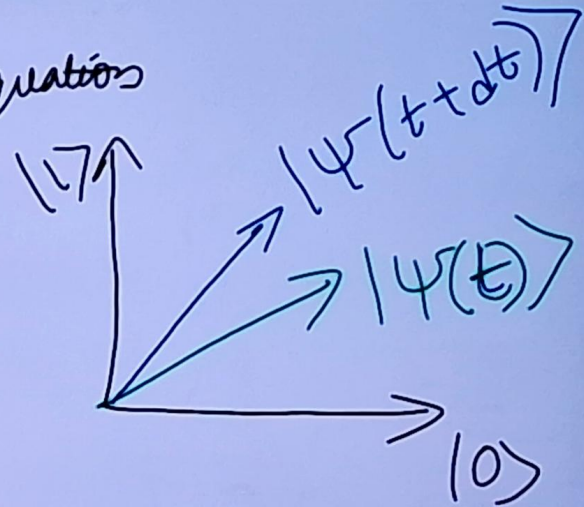


$$i\hbar \frac{d}{dt} |\psi\rangle = \tilde{H} |\psi\rangle$$

$$\tilde{H} = \tilde{T} + \tilde{V}$$

↓ ↓

Kinetic energy operator potential energy operator



Discrete basis

— $\{|i\rangle\}$ for Here $\langle i|j\rangle = \delta_{ij}$

a d -dimensional vector space

Here $\langle x'|x''\rangle = \delta(x'-x'')$

Continuous basis

— specific basis as coordinate basis

$\{|x\rangle\}$ for a ∞ -dimensional vector space

— $|\psi\rangle$ is d -dimensional column vector

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— $\tilde{Q}$ is d -dimensional square matrix

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$$\tilde{I} = \sum_{i=0}^{d-1} |i\rangle\langle i|$$

Kronecker delta function $\delta_{ij} = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases}$

Sampling/Dirac delta function $\delta(x'-x'') = \begin{cases} 0 & \text{if } x' \neq x'' \\ \infty & \text{if } x' = x'' \end{cases}$

$$\tilde{I} = \int_{-\infty}^{+\infty} |x'\rangle\langle x'| dx'$$

Geometrical interpretation of continuous basis representation of $|\psi\rangle$

Consider discrete version

$$|\psi\rangle = \tilde{I} |\psi\rangle$$
$$= \sum_i |i\rangle \underbrace{\langle i|\psi\rangle}_{C_i}$$

Here $C_i := \langle i|\psi\rangle$

$$= \sum_i |i\rangle C_i$$

$$= \sum_i C_i |i\rangle$$

$$\psi(x) = \langle x|\psi\rangle = \int \delta(x-x') \psi(x') dx'$$

Now continuous version

$$|\psi\rangle = \tilde{I} |\psi\rangle$$

$$= \int_{-\infty}^{\infty} |x\rangle \underbrace{\langle x|\psi\rangle}_{\textcircled{A}} dx$$

We define $\langle x|\psi\rangle := \psi(x)$

where $\psi(x)$ is called wavefunction associated with abstract ket $|\psi\rangle$

Left multiply $\textcircled{A}$ with $\langle x|$ gives

$$\langle x|\psi\rangle = \psi(x) = \int_{-\infty}^{\infty} \langle x|x'\rangle \langle x'|\psi\rangle dx'$$

From (A) & (B), we have.

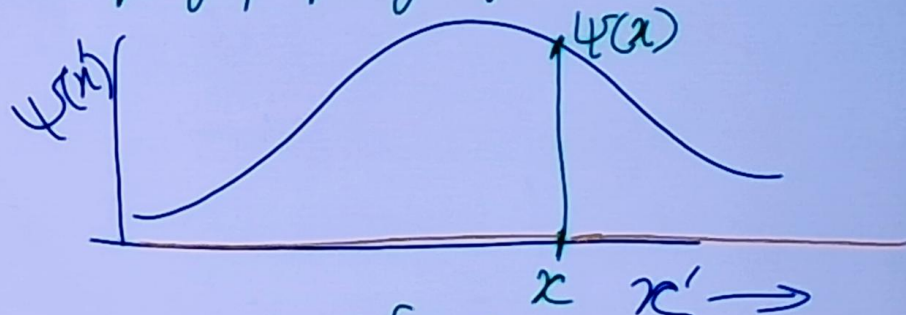
$$\begin{aligned} & \int_{-\infty}^{\infty} \delta(x-x') \psi(x') dx' \\ &= \psi(x) \int_{-\infty}^{\infty} \delta(x-x') dx' \\ &= \psi(x) \end{aligned}$$

$$(B) \quad 1 = \int_{-\infty}^{\infty} \delta(x-x') dx' \leftarrow \delta(x-x') = \begin{cases} 0 & \text{if } x \neq x' \\ \infty & \text{if } x = x' \end{cases}$$

$$(A) \quad \delta(x-x') \cdot \psi(x') = \begin{cases} 0 & \text{if } x \neq x' \\ \psi(x) & \text{if } x = x' \end{cases}$$

$$\psi(x) = \int_{-\infty}^{\infty} \delta(x-x') \psi(x') dx'$$

Sampling property of Dirac delta function



Operators in the coordinate basis

Any observable q is a function of position x and momentum p .

Therefore,

the corresponding operator $\hat{Q}$ will be a function of position operator $\hat{X}$ and momentum operators $\hat{P}$.

— What is the coordinate basis representation of $\hat{X}$?



Consider discrete version.

$$\tilde{Q} = \tilde{I} \tilde{Q} \tilde{I}$$

$$= \left(\sum_i |i\rangle \langle i| \right) \tilde{Q} \left(\sum_j |j\rangle \langle j| \right)$$

$$= \sum_i \sum_j |i\rangle \underbrace{\langle i | \tilde{Q} | j \rangle}_{q_{ij}} \langle j|$$

$$= \sum_i \sum_j \langle i | \tilde{Q} | j \rangle |i\rangle \langle j|$$

$$= \sum_{ij} q_{ij} |i\rangle \langle j|$$

$$\tilde{Q} = \begin{pmatrix} \langle 0 | \\ \langle 1 | \\ \vdots \end{pmatrix} \begin{pmatrix} q_{00} & q_{01} & \dots & q_{0,d-1} \\ q_{10} & q_{11} & \dots & q_{1,d-1} \\ \vdots & \vdots & \ddots & \vdots \\ q_{d-1,0} & q_{d-1,1} & \dots & q_{d-1,d-1} \end{pmatrix}$$

$$= \left(\tilde{Q}|0\rangle \tilde{Q}|1\rangle \dots \tilde{Q}|d-1\rangle \right)$$

$$q_{ij} = \langle i | \tilde{Q} | j \rangle$$



So continuous basis representations of $\tilde{X}$ & $\tilde{P}$

(i) $\tilde{X}$

Action of $\tilde{X}$ on $|x\rangle$

$$\tilde{X}|x\rangle = x|x\rangle$$

$$\tilde{X} = \tilde{I} \tilde{X} \tilde{I}$$

$$= \left(\int_{-\infty}^{\infty} |x'\rangle \langle x'| dx' \right) \tilde{X} \left(\int_{-\infty}^{\infty} |x''\rangle \langle x''| dx'' \right)$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |x'\rangle \underbrace{\langle x' | \tilde{X} | x'' \rangle}_{x''} \langle x''| dx' dx''$$

$$= \iint |x'\rangle \underbrace{x''}_{x'} \langle x' | x'' \rangle \langle x''| dx' dx''$$

$$= \iint |x'\rangle x'' \delta(x' - x'') \langle x''| dx' dx''$$

$$= \int_{-\infty}^{\infty} |x'\rangle x' \langle x'| dx'$$

$$= \int_{-\infty}^{\infty} x' |x'\rangle \langle x'| dx'$$



(ii) $\tilde{p}$



$$\langle x' | \tilde{p} | x'' \rangle = \boxed{\frac{\hbar}{i} \frac{d}{dx'}} \delta(x' - x'')$$



Time Independent Schrodinger equation

1. Discrete version
2. Continuous version

Proof of

(Fri) 27th March, 2026

(A) discrete version of ψ solution to
T. D. S. E.

$$i\hbar \frac{d}{dt} |\psi\rangle = \tilde{H} |\psi\rangle \quad \text{--- (A)}$$

Proof:

Let us assume, we are in $\nearrow$
the energy eigenbasis i.e. $\{ |E_i\rangle \}$
we have solved for the eigenvalues and
eigenkets of $\tilde{H}$.

In this eigenbasis

$$|\psi\rangle = \begin{pmatrix} C_1(t) \\ C_2(t) \\ \vdots \\ C_d(t) \end{pmatrix}$$

at $t=0$

$$|\psi(0)\rangle = \begin{pmatrix} C_1(0) \\ C_2(0) \\ \vdots \\ C_d(0) \end{pmatrix}$$

generally C_i 's
are functions of
time.

then $|\psi(t)\rangle = \begin{pmatrix} C_1(0)e^{-\frac{iE_1t}{\hbar}} \\ C_2(0)e^{-\frac{iE_2t}{\hbar}} \\ \vdots \\ C_d(0)e^{-\frac{iE_d t}{\hbar}} \end{pmatrix}$

$$\tilde{H} = \begin{pmatrix} E_1 & 0 & \dots & 0 \\ 0 & E_2 & & \\ 0 & 0 & & \\ \vdots & \vdots & & \vdots \\ 0 & 0 & & E_d \end{pmatrix}$$

Plugging $|\psi\rangle$ and $\tilde{H}$ in (A)

RHS of (A) is

$$\begin{pmatrix} E_1 C_1(t) \\ E_2 C_2(t) \\ \vdots \\ E_d C_d(t) \end{pmatrix}$$

$$\int f(x) dx = g(x) + \underline{C}$$

LHS of (A) is

$$i\hbar \begin{pmatrix} \frac{d}{dt} C_1(t) \\ \frac{d}{dt} C_2(t) \\ \vdots \\ \frac{d}{dt} C_d(t) \end{pmatrix}$$

LHS = RHS

$$\Leftrightarrow \begin{cases} i\hbar \frac{d}{dt} C_1(t) = E_1 C_1(t) \\ i\hbar \frac{d}{dt} C_2(t) = E_2 C_2(t) \\ \vdots \\ i\hbar \frac{d}{dt} C_d(t) = E_d C_d(t) \end{cases} \text{ set of } d \text{ differential equations}$$

$$\Leftrightarrow \begin{cases} \frac{d}{dt} C_1 = -\frac{iE_1}{\hbar} C_1 \\ \frac{d}{dt} C_2 = -\frac{iE_2}{\hbar} C_2 \\ \vdots \\ \frac{d}{dt} C_d = -\frac{iE_d}{\hbar} C_d \end{cases} \quad \frac{dy}{dx} = \alpha y$$

$$\Leftrightarrow \begin{cases} C_1(t) = C_1(0) e^{-\frac{iE_1 t}{\hbar}} \\ C_2(t) = C_2(0) e^{-\frac{iE_2 t}{\hbar}} \\ \vdots \\ C_d(t) = C_d(0) e^{-\frac{iE_d t}{\hbar}} \end{cases} \quad \frac{dy}{dx} = \beta e^{\alpha x} \cdot \alpha$$

$$\Leftrightarrow \text{LHS of (B) is} \quad \text{RHS of (B)}$$

$$|\psi(t)\rangle = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ \vdots \\ c_d \end{pmatrix} = \begin{pmatrix} c_1(0)e^{-\frac{E_1 t}{\hbar}} \\ c_2(0)e^{-\frac{E_2 t}{\hbar}} \\ \vdots \\ c_d(0)e^{-\frac{E_d t}{\hbar}} \end{pmatrix}$$

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$$|\psi(t)\rangle = \sum$$

$$= c_1(0)e^{-\frac{E_1 t}{\hbar}} \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} + c_2(0)e^{-\frac{E_2 t}{\hbar}} \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix} + \dots + c_d(0)e^{-\frac{E_d t}{\hbar}} \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}$$

$$= c_1(0)e^{-\frac{E_1 t}{\hbar}} |E_1\rangle + c_2(0)e^{-\frac{E_2 t}{\hbar}} |E_2\rangle + \dots + c_d(0)e^{-\frac{E_d t}{\hbar}} |E_d\rangle$$

$$|\psi(t)\rangle = \sum_i c_i(0)e^{-\frac{E_i t}{\hbar}} |E_i\rangle$$

Q.E.D.

quod erat demonstrandum
that which was to be demonstrated

(B) continuous version of solution to

T. D. S.E.

$$\tilde{H}|\psi\rangle = E|\psi\rangle \quad \text{--- (C)}$$

We will work with coordinate basis

$$I = \int_{-\infty}^{\infty} |x'\rangle \langle x'| dx'$$

Using completeness relation

(C) becomes

$$\tilde{H} \left(\int |x'\rangle \langle x'| dx' \right) |\psi\rangle = E|\psi\rangle$$

$$\left(\int_{-\infty}^{\infty} \tilde{H} |x'\rangle dx' \right) \langle x' | \psi \rangle = E |\psi\rangle$$

Take inner product with $\langle x |$ on both sides.

$$\langle x | \left(\int \tilde{H} |x'\rangle dx' \right) \langle x' | \psi \rangle = E \langle x | \psi \rangle$$

$$\int_{-\infty}^{\infty} \langle x | \tilde{H} | x' \rangle dx' \psi(x') = E \psi(x) \quad \text{--- (D)}$$

Now

$$\langle x | \tilde{H} | x' \rangle = \langle x | \tilde{T} + \tilde{V} | x' \rangle$$
$$= \langle x | \frac{\tilde{p}^2}{2m} + V | x' \rangle$$

// Recall $\langle x | \tilde{p} | x' \rangle = \frac{\hbar}{i} \frac{d}{dx} \delta(x-x')$

$$\langle x | \tilde{V} | x' \rangle = V(x) \delta(x-x')$$

Therefore .

$$\langle x | \tilde{H} | x' \rangle = \left(\frac{\hbar}{i} \frac{d}{dx} \frac{\hbar}{i} \frac{d}{dx} + V(x) \right) \delta(x-x')$$

$$= \left(-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x) \right) \delta(x-x')$$

————— (D2)

Plugging (D2) in (D1)

$$\int_{-\infty}^{\infty} \left(-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x) \right) \delta(x-x') \psi(x') dx = E \psi(x)$$

$$\left(-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x) \right) \psi(x) = E \psi(x)$$

$$= \left[-\frac{\hbar^2}{2m} \frac{d^2 \psi}{dx^2} + V(x) \psi(x) = E \psi(x) \right]$$

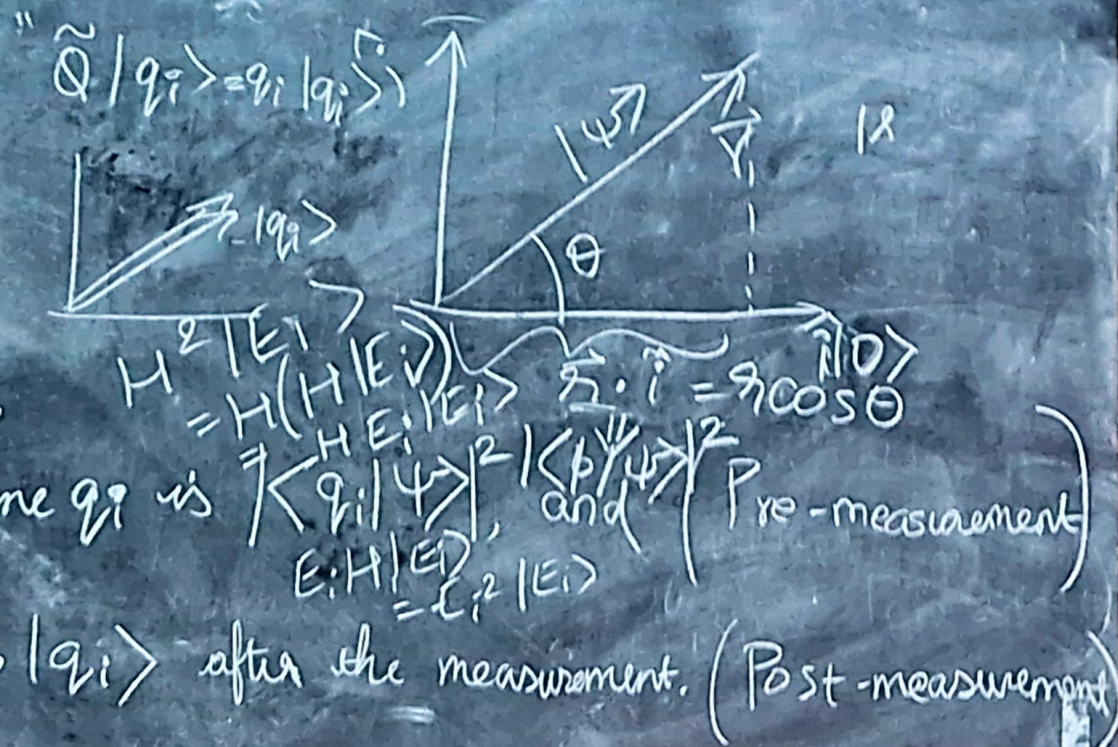
T.I.S.E.

Q. Measurement postulate

Given an observable $\hat{Q}$ and
 given a quantum system in ket $|\psi\rangle$,
 a measurement of $\hat{Q}$ leads to
 outcomes that are eigenvalues $\{q_i\}$.

If the corresponding eigenkets are $|q_i\rangle$,

then (i) the probability of measuring outcome q_i is $|\langle q_i | \psi \rangle|^2$, and (ii) system collapses from $|\psi\rangle$ to $|q_i\rangle$ after the measurement.

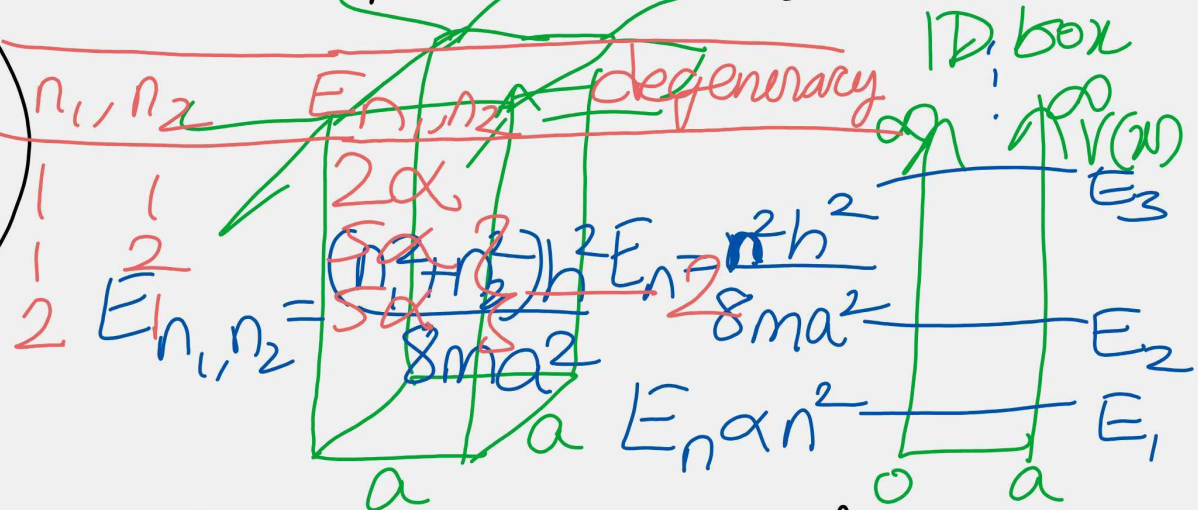


Measurement postulate

$$\tilde{H} = \hbar \omega \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

$$H|E_i\rangle = E_i|E_i\rangle$$

Here computational basis $\{|0\rangle, |1\rangle, |2\rangle\}$



$E_i = \hbar\omega, 2\hbar\omega, 2\hbar\omega$. ; Here there is a case of degeneracy

$$|E_i\rangle = |hw\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad |2hw\rangle = \alpha \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \quad |\alpha|^2 + |\beta|^2 = 1$$

$$\tilde{A} = \lambda \begin{pmatrix} 0 & 1 & | & 0 \\ 1 & 0 & | & 0 \\ \hline 0 & 0 & | & 2 \end{pmatrix}$$

In subspace spanned by $|2\rangle$,

$\tilde{A}$ is diagonal.

$$\tilde{A}_{SS_{|2\rangle}} = (2)_{1 \times 1}$$

$$\Rightarrow A|a_i\rangle = a_i|a_i\rangle$$

so that $a_i = 2\lambda$ and $|2\rangle = (1)_{1 \times 1}$

So in the Hilbert space.

$$|2\lambda\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \quad | \pm \lambda \rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \pm 1 \\ 0 \end{pmatrix}$$

In the subspace spanned by $|0\rangle \otimes |1\rangle$

$$\tilde{A} = \lambda \begin{pmatrix} a(a) & |b) \\ |b) & 0(a) \end{pmatrix}$$

so that $a_i = 1, -1$

$$| \lambda \rangle_{SS_{|0\rangle \otimes |1\rangle}} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}_{2 \times 1}, \quad | -\lambda \rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}_{2 \times 1}$$

$$\tilde{B} = \mu \begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

In subspace spanned by $|0\rangle$

$$\tilde{B} = \mu(2)$$

$$\Rightarrow \tilde{B}|b_i\rangle = b_i|b_i\rangle$$

$$\text{so } b_1 = 2\mu, |2\mu\rangle = (1)$$

Computational basis is $\{|0\rangle, |1\rangle, |2\rangle\}$

In Hilbert space, eigenvectors are

$$|2\mu\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, | \pm \mu \rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ \pm 1 \end{pmatrix}$$

In subspace spanned by $|1\rangle \otimes |2\rangle$

$$\tilde{B} = \mu \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\text{so } b_i = \pm \mu,$$

$$| \mu \rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Expectation value

$$|S(0)\rangle = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = c_1|0\rangle + c_2|1\rangle + c_3|2\rangle \quad \tilde{H} = \hbar\omega \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} \quad \tilde{A} = \lambda \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

$a_i = 2\lambda, \pm\lambda$

$$\langle \tilde{Q} \rangle = \langle \psi | \tilde{Q} | \psi \rangle$$

eigenset of $\hbar\omega = |0\rangle$

$$\Rightarrow \langle 0 | S(0) \rangle = c_1, \quad |a_i\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \quad \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \pm 1 \\ 0 \end{pmatrix}$$

$$\langle \tilde{H} \rangle = \langle \psi | \tilde{H} | \psi \rangle$$

$\Rightarrow \frac{1}{2}(c_1^2 + c_2^2) \Rightarrow |\langle 0 | S(0) \rangle|^2 = |c_1|^2$

$\frac{1}{2}(c_1 + c_2)^2 = \langle \pm\lambda | \psi \rangle^2 = \frac{1}{2} \begin{pmatrix} \pm 1 \\ 1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix}$

$|\psi\rangle = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix}$

$$\langle \tilde{H} \rangle = |c_1|^2 \hbar\omega + (|c_2|^2 + |c_3|^2) 2\hbar\omega$$

$$\langle \tilde{A} \rangle = \frac{1}{2}(c_1^2 + c_2^2 - 2\text{Re}(c_1 c_2^*)) \lambda + \frac{1}{2}(c_1^2 + c_2^2 + 2\text{Re}(c_1 c_2^*)) (\pm\lambda) + |c_3|^2 (+2\lambda)$$

Q. Structures

— Free particle (No structure)

Finite dimensional
Hilbert space



Infinite dimensional
Hilbert space

$$\tilde{Q} = \begin{pmatrix} & \\ & \end{pmatrix}_{d \times d}$$

$$\tilde{Q} = Q(x, p) \quad \hat{p} = \int \frac{\hbar}{i} \frac{d}{dx} \psi(x) dx$$
$$\tilde{X} = \int |x\rangle \langle x| dx$$

Time evolution postulate

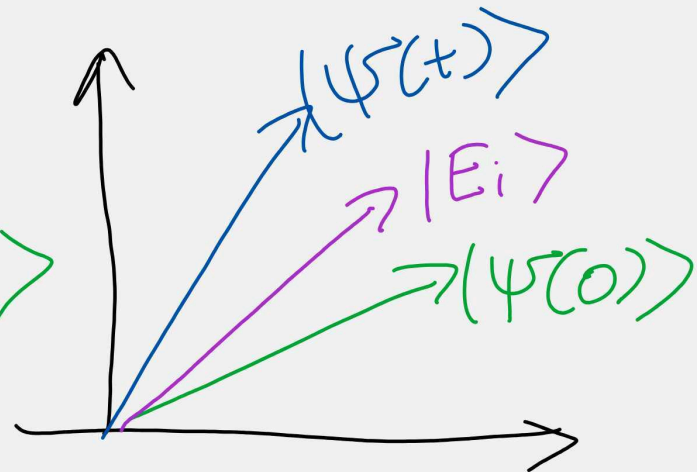
$$\rightarrow \frac{\hbar}{i} \frac{d}{dt} |\psi\rangle = H |\psi\rangle$$

$$|\psi(t)\rangle = e^{-\frac{iHt}{\hbar}} |\psi(0)\rangle$$

Stationary states:

Deterministic states of Hamiltonian.

Deterministic state of $\tilde{Q}$ = any eigenket of $\tilde{Q}$



$$\tilde{Q} = e^{-\frac{i\tilde{H}t}{\hbar}} = 1 + \frac{-iHt}{\hbar} + \frac{1}{2}\left(\frac{-iHt}{\hbar}\right)^2 + \frac{1}{6}\left(\frac{-iHt}{\hbar}\right)^3 + \dots$$

$$\tilde{Q} |q_i\rangle = q_i |q_i\rangle = \left(1 - \frac{it}{\hbar} H - \frac{1}{2}\left(\frac{it}{\hbar}\right)^2 H^2 + \frac{1}{6}\left(\frac{-it}{\hbar}\right)^3 H^3 + \dots\right) |E_i\rangle$$

$|q_i\rangle$ are eigenkets of $\tilde{H}$.

$$\text{Let } H |E_i\rangle = E_i |E_i\rangle$$

Claim:

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \dots = e^{-\frac{it}{\hbar} E_i} |E_i\rangle$$

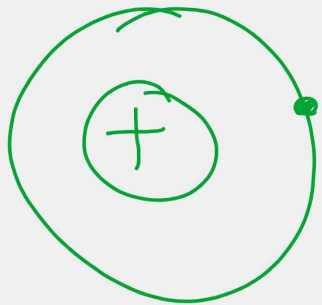
$$= \left(1 + \frac{-it}{\hbar} E_i + \frac{1}{2}\left(\frac{-it}{\hbar}\right)^2 E_i^2 + \frac{1}{6}\left(\frac{-it}{\hbar}\right)^3 E_i^3 + \dots\right) |E_i\rangle$$

$f(x) = f(0) + f'(0)x + f''(0)\frac{x^2}{2} + \dots$

T.E postulate says. $|\psi(t)\rangle = e^{-\frac{itH}{\hbar}} |\psi(0)\rangle$

However, if $|\psi(0)\rangle$ is eigenstate $|E_i\rangle$ of H ,
i.e. $H|E_i\rangle = E_i |E_i\rangle$,

then $|\psi(t)\rangle = e^{-\frac{iHt}{\hbar}} |E_i\rangle$



$$|\psi(t)\rangle = \underbrace{e^{-\frac{iE_i t}{\hbar}}}_{e^{i\delta}} |E_i\rangle$$

$|n=1, l=0, m=0, s=\frac{1}{2}\rangle$

$$|\psi'\rangle = e^{i\delta} |\psi\rangle$$

$$\langle \psi' | \psi' \rangle = 1$$

Waveket $|\psi\rangle$

Finite

$$H |E_i\rangle = E_i |E_i\rangle$$

$$\left(\begin{array}{c} \\ \\ \\ \end{array} \right)_{d \times d}$$



vs

Wavefunction $\langle x|\psi\rangle \equiv \psi(x)$

Infinite

$$\tilde{H} = \tilde{T} + \tilde{V}$$

In coordinate basis representation

$$\begin{array}{ccc} \tilde{H} & = & \tilde{T} + \tilde{V} \\ \downarrow & & \downarrow \quad \downarrow \\ H(x) & = & \frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x) \end{array}$$

$$H(x) E_i(x) = E_i E_i(x)$$